Giovanni Sullutrone

LLM safety researcher · refusal, over-refusal and safety evaluation

Modena, Italy · giovanni.sullutrone@unimore.it · +39 388 351 4673 · giosullutrone.github.io · GitHub · Google Scholar · LinkedIn

I build evaluations of when large language models refuse, ask for clarification or abstain. My first-author work measures over-refusal that retrieved documents trigger (COVER, Findings of ACL 2025), refusal on database schemas with personal and sensitive data, and clarification in multi-turn interaction with simulated users (ABISS). I am looking for research scientist, research engineer and postdoctoral roles in LLM safety, evaluation and interpretability.

Education

PhD in Information and Communication Technologies, *University of Modena and Reggio Emilia (UNIMORE)* Nov 2023 – Oct 2026

DBGroup, Department of Engineering "Enzo Ferrari". Supervisor: Prof. Sonia Bergamaschi. The PhD cycle ends in October 2026, and the defence is expected in early 2027.

M.Sc. in Computer Engineering, *University of Salerno* Oct 2021 – Oct 2023

Artificial Intelligence and Intelligent Robotics track. Graduated 110/110 cum laude.

B.Sc. in Computer Engineering, *University of Salerno* Oct 2018 – Oct 2021

Graduated 110/110.

Research experience

PhD Researcher, LLM safety and evaluation, *UNIMORE, Modena, Italy* Nov 2023 – Oct 2026

- Led first-author studies of refusal and over-refusal in retrieval-augmented generation and Text-to-SQL, within the ITSERR research infrastructure project (PNRR, NextGenerationEU).
- Built evaluation pipelines for open-weight and API models with vLLM, across question answering, translation, summarization and SQL generation.
- Contributed to OCR, post-OCR and cataloguing research for Arabic-script collections in the Digital Maktaba work package.

Visiting Researcher, *DARE Lab, Athena Research Center, Athens, Greece* (host: Dr. Georgia Koutrika) Apr – Jul 2025

- Designed ABISS, a benchmark that evaluates Text-to-SQL agents on ambiguous and unanswerable requests through multi-turn interaction with simulated users.

Machine Learning Researcher (research grant), *A.I. Tech S.R.L., Fisciano, Italy* 2023

- Defined user requirements and designed and implemented a monitoring and prediction system for public transport management.

Selected publications (first author)

COVER: Context-Driven Over-Refusal Verification in LLMs Findings of ACL 2025

G. Sullutrone, R. A. Vigliermo, S. Bergamaschi, L. Sala paper code

Introduces context-driven over-refusal: a model rejects a benign task because of the supplied documents, not the request. Translation and summarization of the same evidence were refused far more often than question answering, in some settings for every request. Later work cites COVER in blocking attacks on RAG systems and in hidden-state monitoring of harmful dialogue.

Text-to-Refused-SQL: A Comprehensive Evaluation of LLMs Refusal in Text-to-SQL SEBD 2025

G. Sullutrone, L. Sala, L. Trigiante, S. Bergamaschi paper

Measures schema-driven over-refusal on healthcare databases with personal and sensitive fields. Affected models refused non-sensitive questions because sensitive columns appeared elsewhere in the schema. Stating that the user was authorized more than doubled refusal in almost every affected setting.

ABISS: Evaluating Text-to-SQL Systems Through Agent Interaction arXiv 2026 · committed to EACL 2027

G. Sullutrone, L. Sala, S. Aftar, G. Koutrika, S. Bergamaschi paper

Unifies the reasons why a database question can be ambiguous or unanswerable. A multi-agent pipeline generates and validates 7,145 questions on BIRD and Spider, and simulated users test the full interaction. Agents often noticed that a request was problematic without identifying why, and struggled to turn useful clarification into correct SQL.

Sensitive Topics Retrieval in Digital Libraries: A Case Study of ḥadīṯ Collections TPDL 2024

G. Sullutrone, R. A. Vigliermo, L. Sala, S. Bergamaschi paper

Tests ten embedding models on about 12,000 questions generated from 4,000 passages. Sensitive passages were usually retrieved but ranked lower than non-sensitive ones, with up to 26% ranking degradation, while BM25 stayed stable.

Other publications

1. "Text-to-SQL with Large Language Models: Challenges Revisited and New Dimensions." L. Sala, **G. Sullutrone**, S. Bergamaschi. *DATA 2026*, position paper.
2. "The 'Digital Maktaba LP': Proposing a Comprehensive Dataset for Arabic Script OCR Title Pages." R. A. Vigliermo, **G. Sullutrone**, S. Bergamaschi, L. Sala. *Umanistica Digitale* 22, 2026, journal article.
3. "The Digital Maktaba Platform: A Tool for Semi-Automatic Document Cataloguing and Searching." S. Bergamaschi, F. Ruozzi, D. Ciccarello, R. A. Vigliermo, L. Sala, A. El Ganadi, **G. Sullutrone**, S. Aftar. *The Digital Turn in Religious Studies*, V&R unipress, 2025, book chapter.
4. "DM-LP: A Large Arabic Script Languages Dataset for OCR and Cataloguing Research." L. Sala, R. A. Vigliermo, **G. Sullutrone**, S. Bergamaschi. *TPDL 2025*, short paper.
5. "Proposing a Comprehensive Dataset for Arabic Script OCR in the Context of Digital Libraries and Religious Archives." R. A. Vigliermo, **G. Sullutrone**, S. Bergamaschi, L. Sala. *IRCDL 2025*, extended abstract.
6. "Arabic Text Enhancement with GPT for Digital Libraries." L. Sala, **G. Sullutrone**, S. Bergamaschi, R. A. Vigliermo. *JCDL 2024*, short paper. Second place, Best Short Paper Award.
7. "Text-to-SQL with Large Language Models: Exploring the Promise and Pitfalls." L. Sala, **G. Sullutrone**, S. Bergamaschi. *SEBD 2024*.
8. "Large Language Models Integration in Digital Humanities." **G. Sullutrone**. *SEBD 2024*, doctoral consortium.

Research projects

ITSERR, *PNRR research infrastructure project, NextGenerationEU* (Digital Maktaba work package) Nov 2023 – Oct 2026

EpiCDIn, *Emilia-Romagna Region* (PI: Prof. Rossana Cecchi, with the University of Copenhagen) Dec 2025 – Dec 2026
Participant in a feasibility project on AI-guided epigenetic approaches to cardiovascular disease and sudden unexplained death.

Open-source code

COVER · evaluation pipeline for context-driven over-refusal, with vLLM, OpenAI and Gemini back-ends github.com/giosullutrone/COVER

Other experience

Software Development Consultant (freelance), *Datariver* Mar – Jul 2026
Research software and data-handling tools for clinical research and healthcare data management, with testing and technical documentation.

Co-founder, *Vivi S.R.L.* since 2023
Startup that develops a device to serve wine by the glass from sealed bottles.

Awards

Second place, Best Short Paper Award, JCDL 2024, for "Arabic Text Enhancement with GPT for Digital Libraries" (second author) Dec 2024

Skills

Research	LLM safety evaluation, refusal and over-refusal, benchmark design, multi-turn and agent evaluation with simulated users, retrieval-augmented generation, Text-to-SQL, synthetic data with multi-agent validation
Engineering	Python, PyTorch, Hugging Face Transformers, vLLM, TRL, LoRA fine-tuning, Docker, SQL, Git
Other	OCR and post-OCR correction for Arabic-script documents

All twelve publications with links: giosullutrone.github.io · Updated October 2026